IGCSE 0607 Practice Questions — E1.9

Estimation

Total Questions: 18 — **Total Marks:** 90 Marks

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1 Consider the number 47.5381.

(a) Write 47.5381 correct to:

(i) 2 decimal places.

..... **[1]**

(ii) 2 significant figures.

..... **[1]**

(iii) the nearest integer.

..... **[1]**

2 Write the following numbers correct to the specified degree of accuracy.

(a) 782.51 correct to the nearest ten.

..... **[1]**

(b) 0.03561 correct to 2 significant figures.

..... **[1]**

(c) 4.098 correct to 2 decimal places.

..... **[1]**

3 **(a)** Write 48.975 correct to 2 decimal places.

..... **[1]**

(b) Write 0.003058 correct to 2 significant figures.

..... **[1]**

(c) By writing each number correct to 1 significant figure, estimate the value of

$$\frac{7.84 \times 31.6}{0.385}$$

..... **[2]**

4 **(a)** Round the following numbers to the specified degree of accuracy.

(i) 34.567 correct to 2 decimal places.

..... **[1]**

(ii) 0.070381 correct to 2 significant figures.

..... **[1]**

(b) By writing each number correct to 1 significant figure, estimate the value of

$$\frac{61.3 \times 3.92}{0.485}$$

..... **[2]**

5 **(a)** Write 0.04758 correct to 2 significant figures.

..... **[1]**

(b) Write 384.6 correct to the nearest 10.

..... **[1]**

(c) By writing each number correct to 1 significant figure, estimate the value of

$$\frac{58.7 \times 4.12}{19.6 - 2.87}(0)$$

..... **[2]**

6 **(a)** Write 203.498 correct to

(i) 2 decimal places.

..... **[1]**

(ii) 3 significant figures.

..... **[1]**

(iii) the nearest ten.

..... **[1]**

(b) By writing each number correct to 1 significant figure, estimate the value of

$$\frac{6.89 \times 41.2}{0.513}$$

..... **[2]**

7 By writing each number correct to 1 significant figure, estimate the value of

$$\frac{58.4 \times 3.09}{0.187}$$

..... **[3]**

8 **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{7.84 \times 31.6}{0.381}$$

..... **[2]**

(b) By writing each number correct to 1 significant figure, estimate the value of

$$\frac{\sqrt{98.6} + 19.3}{0.513^2}$$

..... **[3]**

9 **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{\sqrt{98.5} \times 4.12^2}{0.187}$$

..... **[3]**

(b) A rectangular field has length 489 m and width 215 m.

Estimate the area of the field, in square metres, by rounding each length to 1 significant figure.

..... m^2 **[2]**

10 **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{\sqrt{3.87 \times 10^6}}{0.0524 \times 21.3}$$

..... **[3]**

(b) Is your estimate in part (a) greater than or less than the exact value of the expression? Give a reason for your answer.

.....

..... **[2]**

11 **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{\sqrt[3]{8.25} \times 19.3^2}{0.485}$$

..... **[3]**

(b) The number of bacteria in an experiment is given by the formula

$$N = 4.12 \times 10^5 \times (1.95)^t$$

where t is the time in hours.
By writing each base number correct to 1 significant figure, estimate the number of bacteria when $t = 3$.
Give your answer in standard form.

$N \approx$ ..... **[3]**

12 **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{\sqrt{98.5} \times 4.1^3}{0.192 \times 48.7}$$

You must show all your working.

..... **[3]**

(b) Given that $y = \dfrac{ax}{b-c}$, find an estimate for the value of y when
$a = 3.92$, $x = 0.0514$, $b = 1.08$ and $c = 0.812$.
Write all numbers correct to 1 significant figure in your calculation.

$y =$ ..... **[3]**

13 **(a)** By writing each number correct to 1 significant figure, estimate the value of

$$\frac{38.7 \times 5.12^2}{\sqrt[3]{996}}$$

..... **[3]**

(b) A solid metal cylinder has radius 3.89 cm and height 11.2 cm.

By writing each dimension correct to 1 significant figure, estimate the volume of the cylinder.
Take the value of π to be 3.

..... cm^3 **[3]**

(c) A train travels a distance of 4210 km at an average speed of 585 km/h.

By writing both the distance and the speed correct to 1 significant figure, estimate the time taken for the journey.
Give your answer in hours and minutes.

hours, minutes **[3]**

14 A rectangular field has length 125 m and width 74 m.

(a) Calculate the area of the field.
Give your answer correct to 2 significant figures.

..... m^2 **[2]**

(b) A different field is a square of side x metres.
The area of this square field is equal to the area of the rectangular field.
Calculate the value of x.
Give your answer correct to 1 decimal place.

$x =$ ..... **[3]**

15 A solid metal cone has radius $r = 5.82$ cm and height $h = 11.4$ cm.

(a) Calculate the volume of the cone.
Give your answer correct to 3 significant figures.
The volume, V, of a cone with radius r and height h is $V = \frac{1}{3}\pi r^2 h$.

..... cm^3 **[2]**

(b) By writing each of the values 5.82, 11.4 and π correct to 1 significant figure, estimate the volume of the cone.
Show your working.

..... cm^3 **[2]**

(c) A worker packs these cones into boxes.
Each box holds exactly 12 cones.
The factory produces 850 cones a day.
Calculate the maximum number of boxes that can be completely filled each day.

..... **[2]**

16 **(a)** Calculate the value of

$$\frac{4.2^3 + \sqrt{18.5}}{0.75 \times 1.2}$$

Give your answer correct to 3 significant figures.

..... **[2]**

(b) A spherical water tank has a radius of 1.85 metres.

(i) Calculate the volume of the tank.
Give your answer correct to the nearest cubic metre.
The volume, V, of a sphere with radius r is $V = \frac{4}{3}\pi r^3$.

..... m^3 **[2]**

(ii) The tank is initially empty.
Water is pumped into the tank at a rate of 12.5 litres per minute.
Calculate the time required to fill the tank completely.
Give your answer in hours and minutes, correct to the nearest minute.

hours, minutes **[3]**

17 **(a)** Calculate the value of

$$p = \frac{28.7^2 - 11.2^2}{\sqrt{98.5}}$$

(i) Write down all the figures on your calculator display.

$p =$ ..... **[1]**

(ii) Write your answer to part (a)(i) correct to 3 significant figures.

$p =$ ..... **[1]**

(b) By writing each number in the original expression correct to 1 significant figure, estimate the value of p.

$p \approx$ ..... **[2]**

18 A solid metal cylinder has a base radius of 4.15 cm and a height of 19.8 cm. The metal has a density of 8.13 grams per cubic centimetre.

(a) By writing each number correct to 1 significant figure, estimate the mass of the cylinder in grams.
Use $\pi \approx 3$ for your estimation.
Show your working clearly.

..... g **[3]**

(b) Calculate the actual mass of the cylinder.
Give your answer correct to 3 significant figures.

..... g **[3]**